

Rahul Kumar Singh

AI & Data Science Engineer

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Summary

AI & Data Science undergraduate passionate about Machine Learning, Deep Learning, Computer Vision, and Full-Stack AI systems. Experienced in building and deploying end-to-end AI applications using YOLOv8, Flask, React.js, and MongoDB. Skilled in EDA, feature engineering, model evaluation, API integration, and cloud deployment for real-world AI solutions.

Skills

Languages : Python, C++, JavaScript, SQL
ML / AI : Scikit-learn, YOLOv8, TensorFlow, OpenCV, Pandas, NumPy, Matplotlib, Seaborn
Full-Stack : React.js, Node.js, Express.js, Flask, MongoDB Atlas, REST APIs
Tools : Git, GitHub, Streamlit, Jupyter, Vercel, Render, Hugging Face Spaces
Concepts : Machine Learning, Deep Learning, Computer Vision, EDA, Feature Engineering, Full-Stack AI Deployment

Work Experience

Machine Learning Intern

Feb 2026 – May 2026

RVNS Solutions Pvt. Ltd.

Remote

- Designed and deployed an **AI-Based Oral Health Analysis System** using **YOLOv8** capable of detecting 5 oral diseases from mouth images with real-time color-coded annotations.
- Built full-stack architecture using **React.js**, **Node.js**, **Express.js**, **Flask API**, **MongoDB Atlas** with authentication, prediction history dashboard, and downloadable reports.
- Integrated custom-trained YOLOv8 model with Flask API and deployed the complete system on **Vercel**, **Render**, and **Hugging Face Spaces** for cloud-based real-time inference.

Machine Learning Intern

Jan 2025 – Feb 2025

CodeAlpha

Remote

- Built ML models for **Credit Scoring** (multi-class) and **Breast Cancer prediction** (binary) using Logistic Regression, Random Forest, and SVM on real-world datasets.
- Evaluated performance using Accuracy, F1-score, and ROC-AUC; achieved **~98% accuracy** across tasks.
- Delivered end-to-end ML workflows using Python and Scikit-learn from raw data to model evaluation.

Projects

AI-Based Oral Health Analysis System — YOLOv8, Full-Stack AI Healthcare 🌐 [Live Demo](#) 🐙 [GitHub](#) 2026

- Built and deployed a full-stack AI healthcare app detecting 5 oral diseases (Calculus, Caries, Gingivitis, Ulcer) using a custom-trained **YOLOv8** model with **React.js**, **Node.js**, **Express.js**, **Flask**, and **MongoDB Atlas**.
- Implemented real-time image inference, color-coded disease annotations, prediction history tracking, authentication, and downloadable reports; deployed on **Vercel**, **Render**, and **Hugging Face Spaces**.

Customer Churn Prediction System — End-to-End ML, Deployed 🐙 [GitHub](#) 2026

- Built churn prediction on Telco data (7k+ rows, 26 features) using **Scikit-learn Pipelines** and **ColumnTransformer**; handled data leakage, missing values, and multi-source merging.
- Trained Logistic Regression, Random Forest, Gradient Boosting; tuned LR achieved **ROC-AUC $\approx$ 0.84**; deployed as **Streamlit** app with real-time churn probability and risk explanation.

IPL 50+ Score Prediction System — EDA $\rightarrow$ ML $\rightarrow$ Deployment 🐙 [GitHub](#) 2026

- Converted IPL batting dataset (15k+ rows) into a classification problem; performed EDA, feature engineering, and data leakage prevention.
- Final Gradient Boosting model achieved **F1 $\approx$ 0.80**, **ROC-AUC $\approx$ 0.98**; deployed via **Streamlit** for real-time predictions

Drone Surveillance System for Anomaly Detection — AI Hackathon Project 🐙 [GitHub](#) 2025

- Designed an AI-based drone monitoring system for fire, flood, and anomaly detection using ML algorithms.
- Built a real-time simulation and analytics pipeline using **FastAPI** and **PyBullet** for drone environment modeling.

Education

Dr. D. Y. Patil Institute of Engineering, Management & Research (DYPIEMR)

2023 – 2027

B.E. in Artificial Intelligence & Data Science, Pune

CGPA: 8.55

Certifications

- Complete Data Science, ML, Deep Learning & NLP Bootcamp** (99 Hours) — Udemy
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** — Oracle
- Python: From Zero to Hero & Complete MySQL Bootcamp** — Udemy